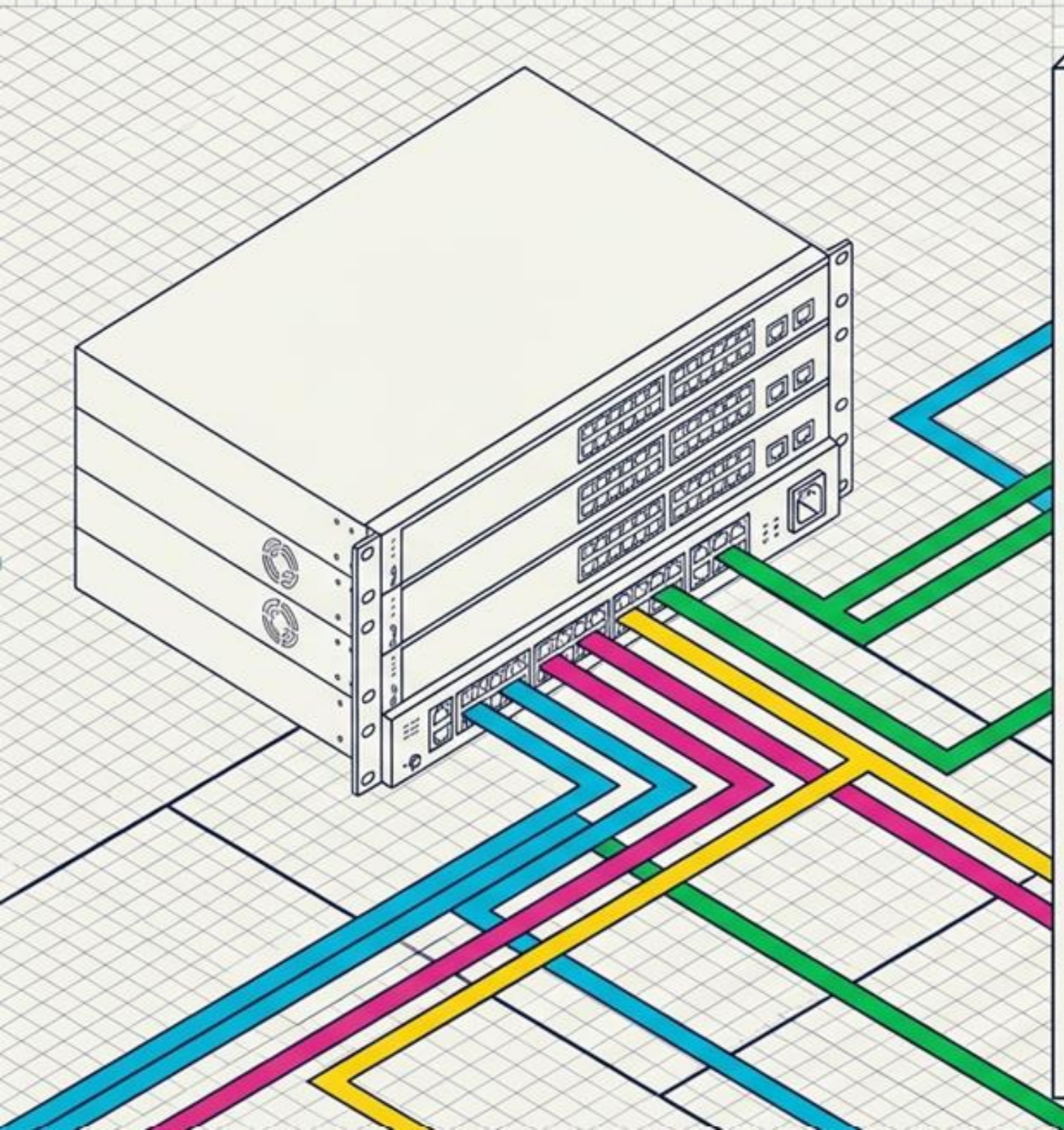


ARCHITECTING SCALABLE INTER-VLAN ROUTING FOR MCNASIR SYSTEMS

The Evolutionary Migration from Traditional Routing to Layer 3 Multilayer Switching.

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Executive Summary: Network Architecture Evolution

Chevron 1



The Driver

Three years of steady digital and staff growth necessitated expanding departmental network segmentation.

Chevron 2



The Challenge

Connecting isolated departments (VLANs) rapidly depleted physical router ports and switch uplinks.

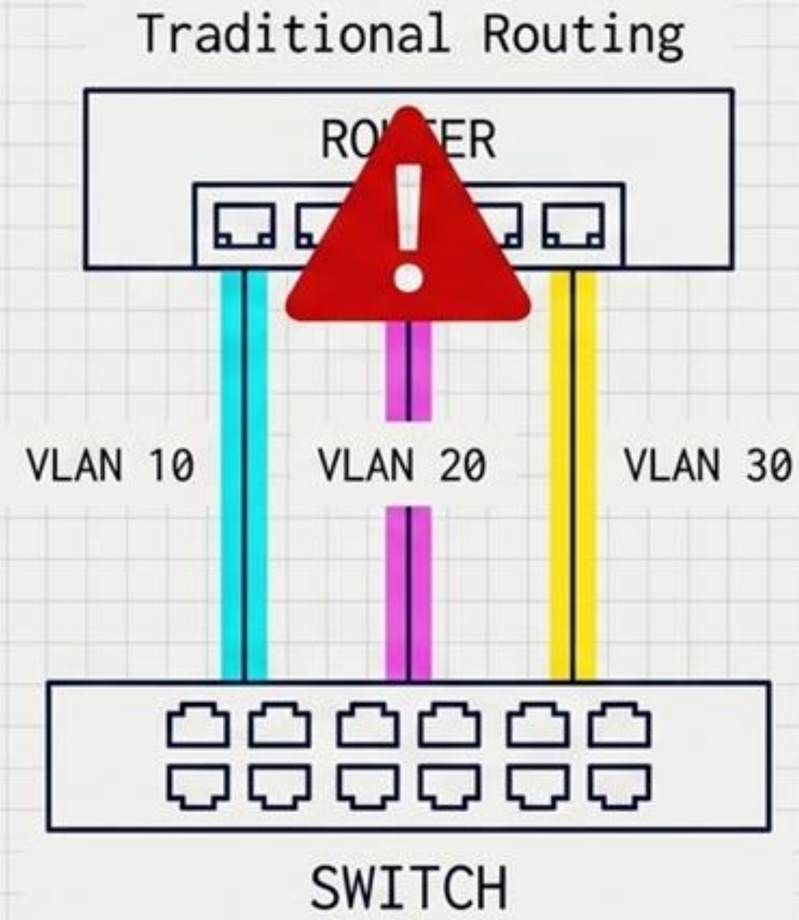
Chevron 3



The Solution

Migrating inter-VLAN routing entirely to hardware-speed Layer 3 switching via Switch Virtual Interfaces (SVIs), eliminating external router dependencies.

The Scalability Bottleneck of Traditional Routing



Key Insight

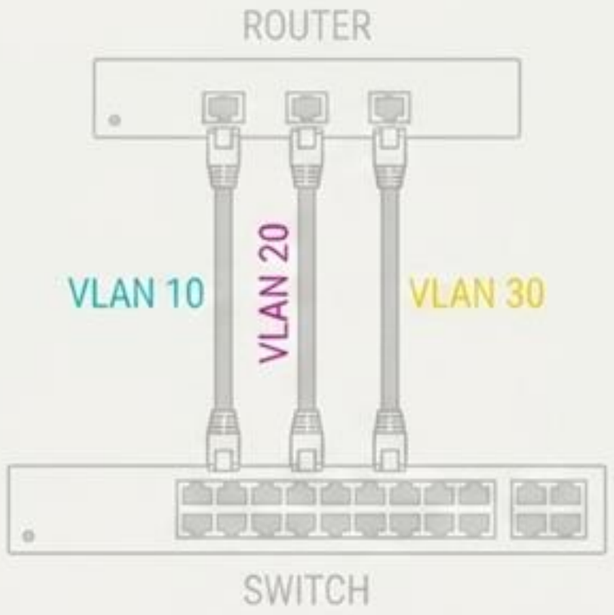
The traditional method required one dedicated physical router interface and one physical cable per VLAN.

The Business Impact

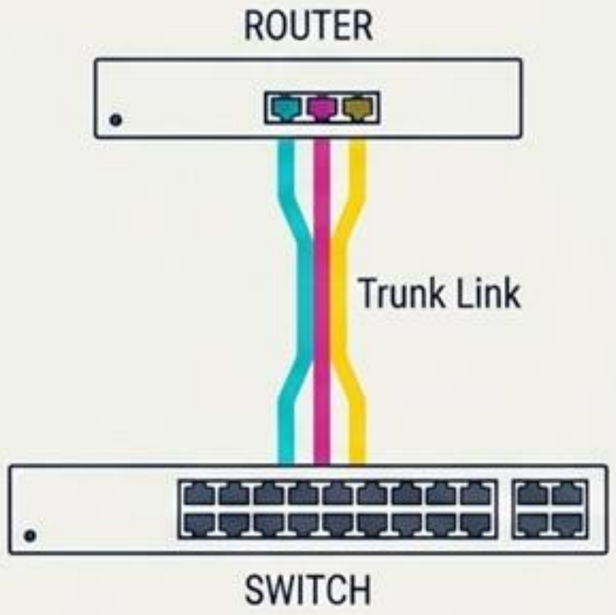
As McNasir Systems grew, adding new departments meant physically maxing out router ports and switch uplinks, making horizontal expansion cost-prohibitive and physically unsustainable.

Phase 2 Network Expansion via Router-on-a-Stick (RoaS)

Traditional (Faded)



RoaS (Active)



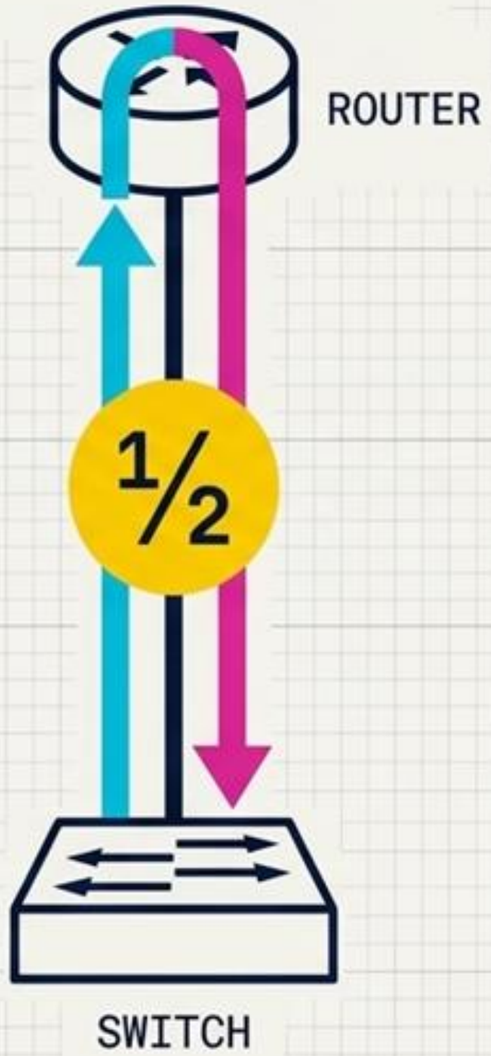
Mechanics

To support the addition of VLAN 30 (Finance) and the reassignment of VLAN 20 (Marketing), the network migrated to RoaS.

Configuration Data

A single physical router interface (Gi0/0) leverages 802.1Q sub-interfaces (.10, .20, .30) to route traffic over a single physical trunk cable, freeing up physical switch ports.

The Inherent Limitations of the RoaS Architecture



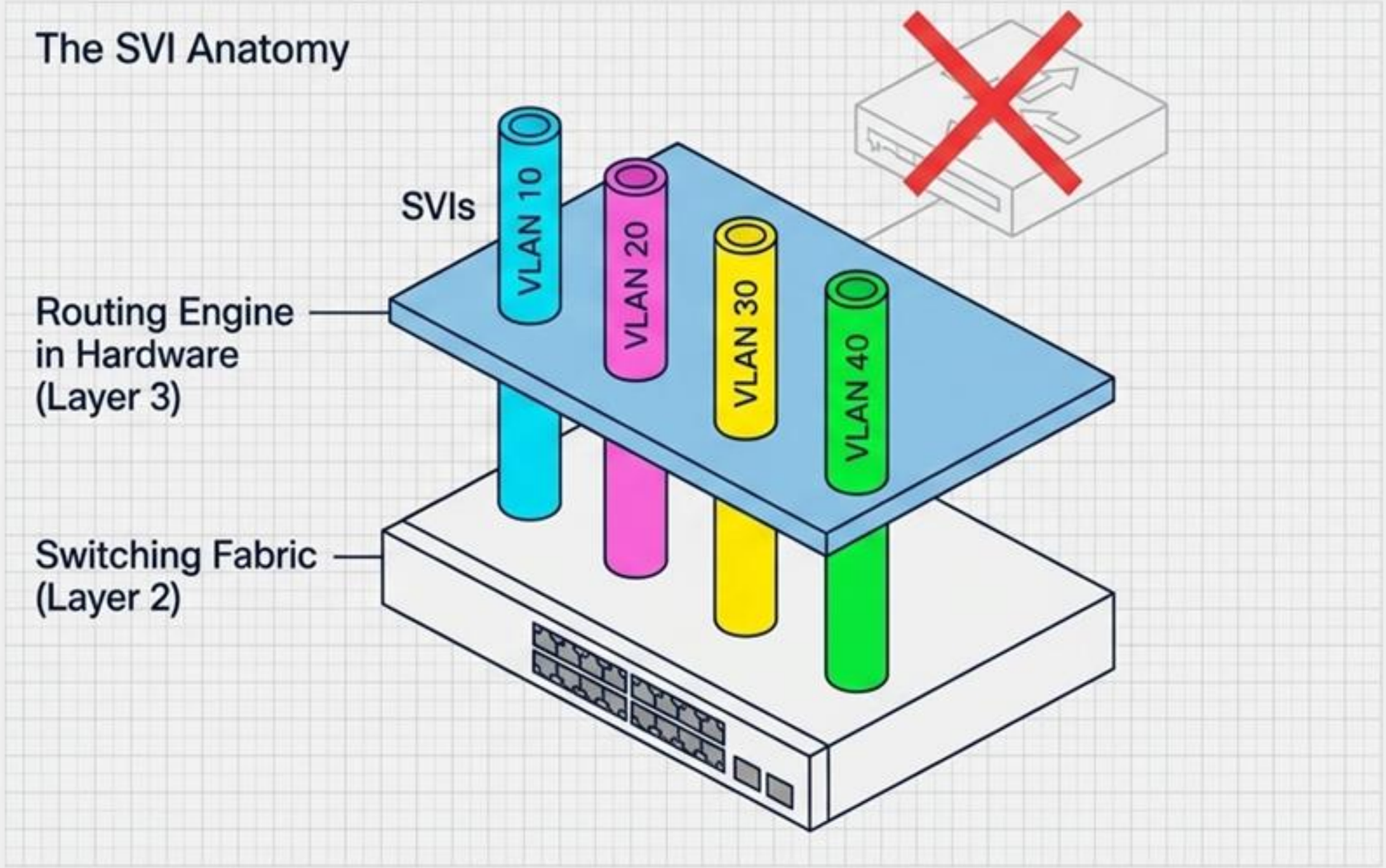
Constraint 1 - Hardware Dependency

The network remains entirely dependent on external routing hardware to process inter-departmental traffic.

Constraint 2 - The Bandwidth Bottleneck

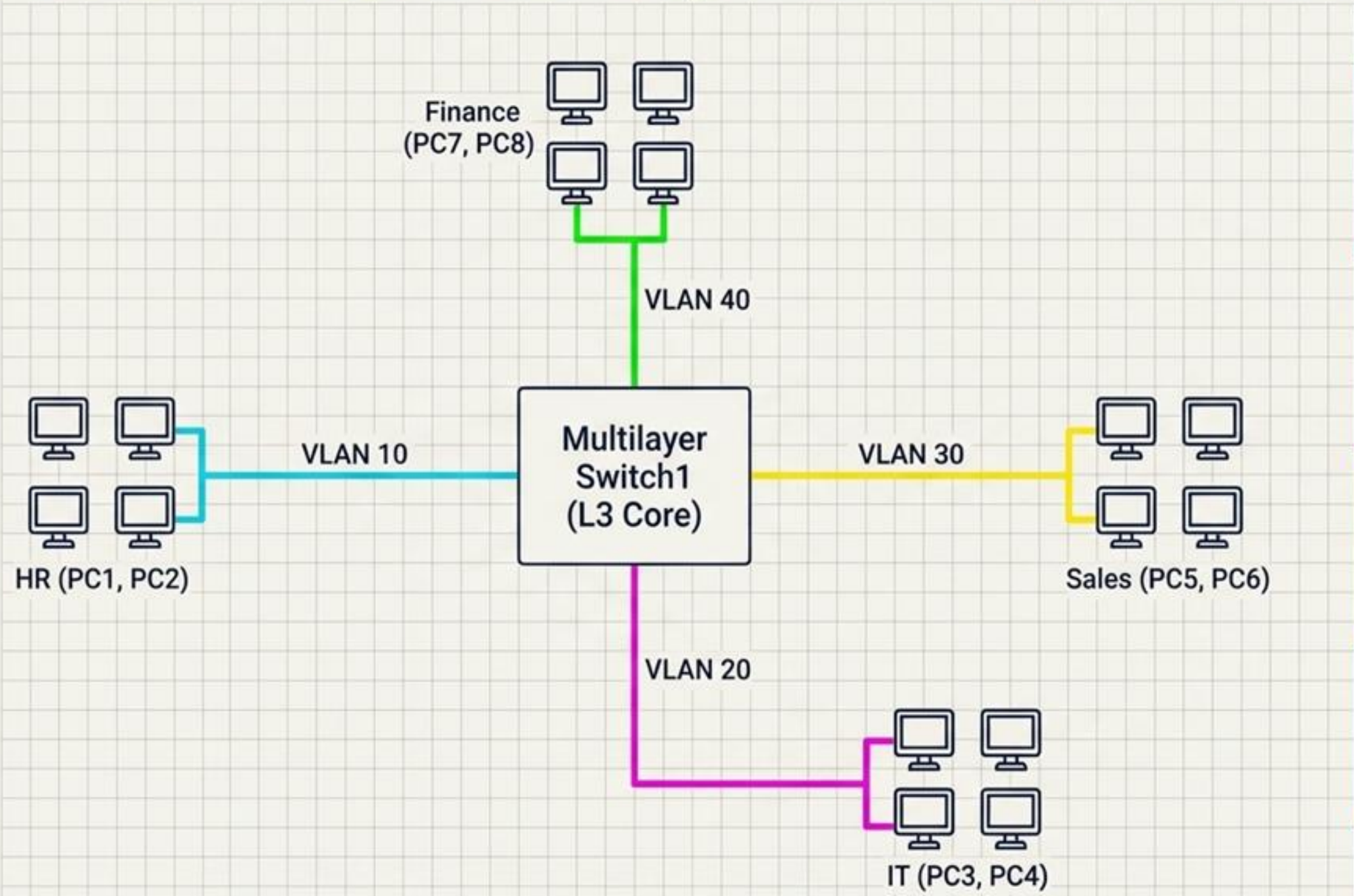
All inter-VLAN traffic must traverse the single physical trunk link twice (up and down). This 'hairpin' routing effectively halves the available throughput, severely degrading performance during peak enterprise loads.

Phase 3 Introduces the Layer 3 Multilayer Switch



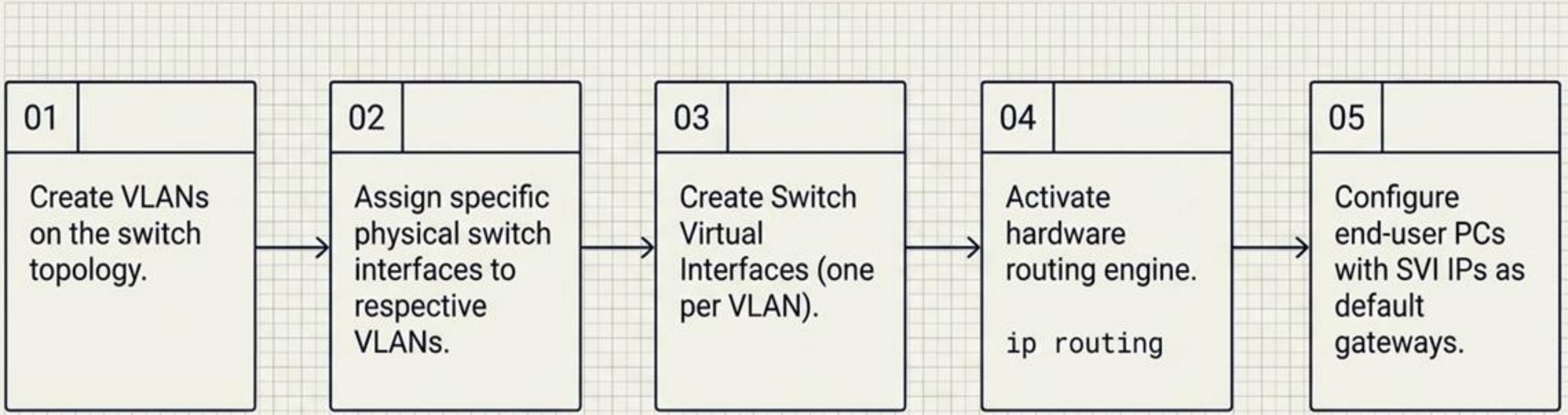
1	Core Concept
	By creating Switch Virtual Interfaces (SVIs), routing happens internally in hardware at wire speed.
2	The Breakthrough
	The external router and the congested trunk link are completely eliminated. Inter-VLAN routing is now a purely logical function.

The Finalized McNasir Systems Enterprise Core



Legend	
	VLAN 10 (HR)
	Net: 192.168.1.0/24 | SVI GW: 192.168.1.1
	VLAN 20 (IT)
	Net: 192.168.2.0/24 | SVI GW: 192.168.2.1
	VLAN 30 (Sales)
	Net: 192.168.3.0/24 | SVI GW: 192.168.3.1
	VLAN 40 (Finance)
	Net: 192.168.4.0/24 | SVI GW: 192.168.4.1

The Layer 3 Implementation Playbook



Takeaway: Adding a new department is now a highly scalable, purely logical configuration requiring zero new cabling.

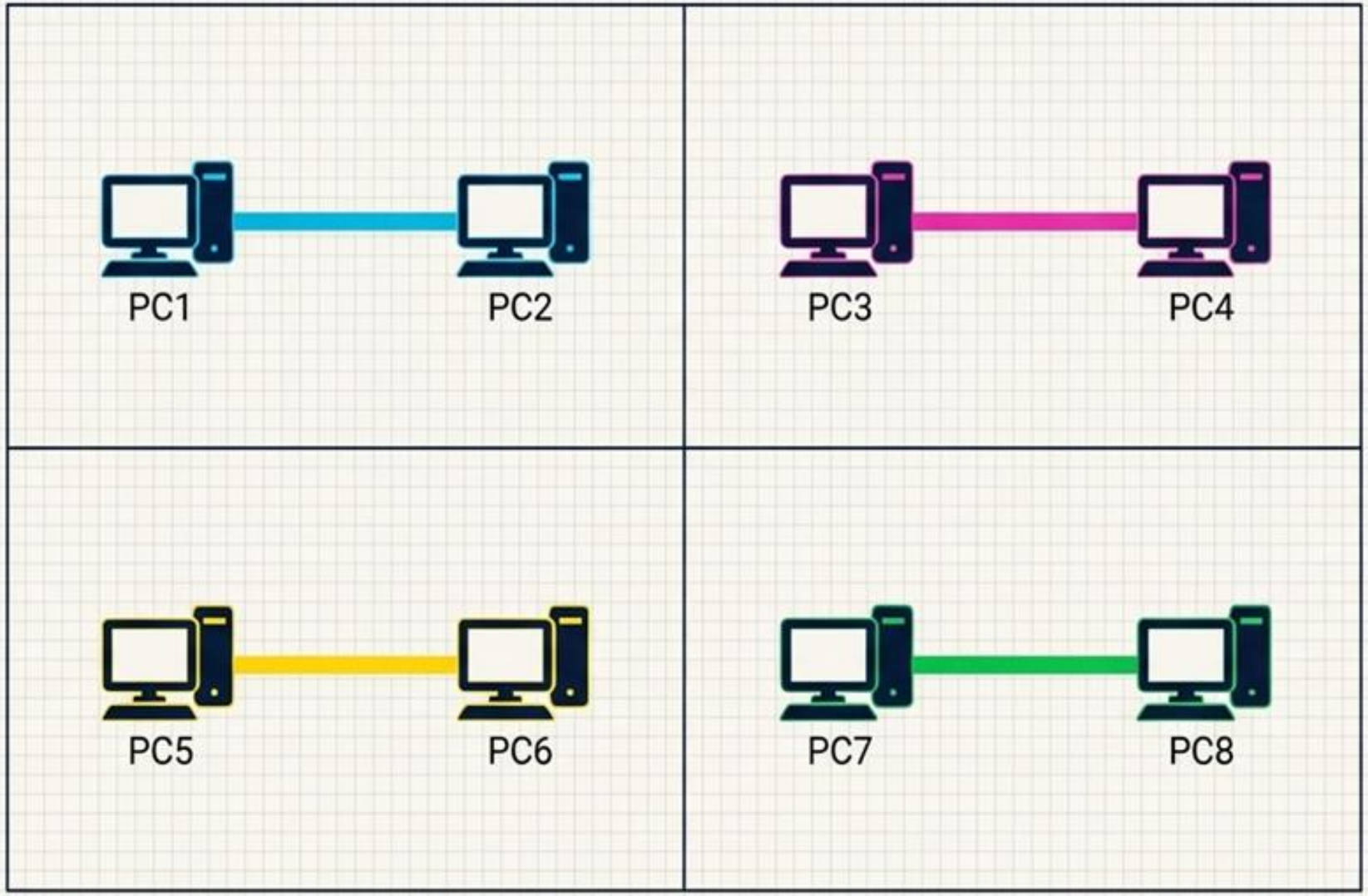
Under the Hood: System Verification Dashboard

VLAN Active Status (Derived from show vlan brief)	
	VLAN 10 (HR) | Active | Gig1/0/1–2
	VLAN 20 (IT) | Active | Gig1/0/3–4
	VLAN 30 (Sales) | Active | Gig1/0/5–6
	VLAN 40 (Finance) | Active | Gig1/0/7–8

Routing Engine Status (Derived from show ip route)	
C	192.168.1.0/24 is directly connected, Vlan10
C	192.168.2.0/24 is directly connected, Vlan20
C	192.168.3.0/24 is directly connected, Vlan30
C	192.168.4.0/24 is directly connected, Vlan40

The routing table proves the switch is successfully aware of all subnets internally.

Baseline Validation: Intra-VLAN Layer 2 Connectivity



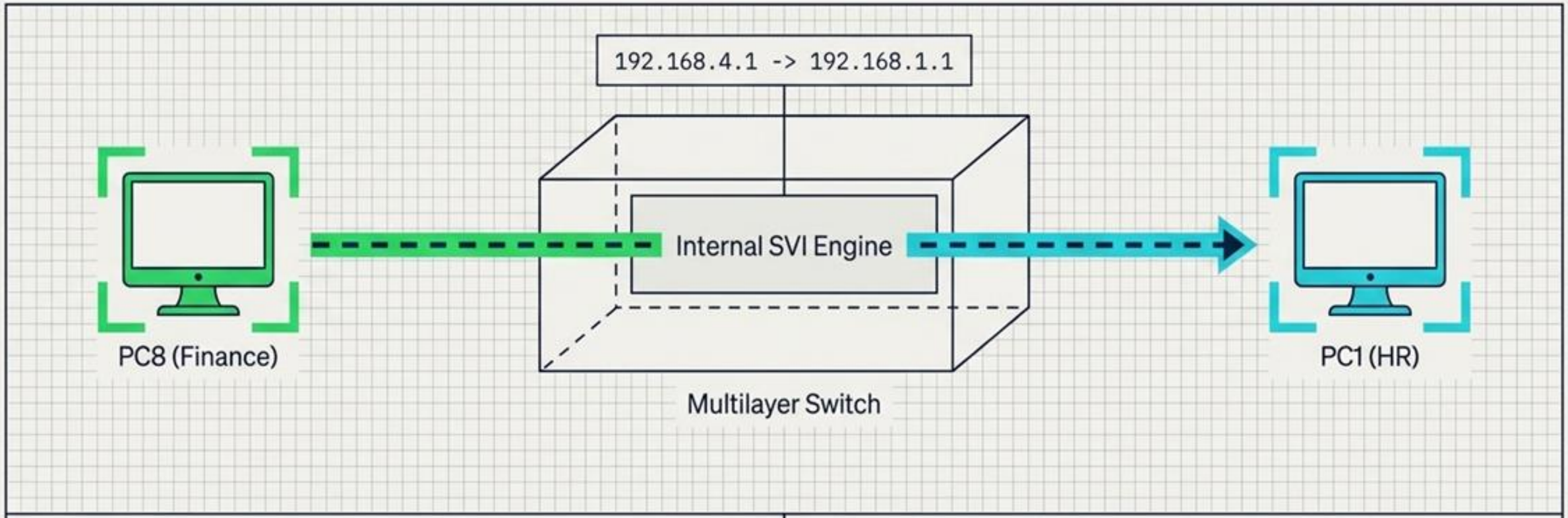
Observation

Pings sent between PCs within the same department (e.g., PC1 to PC2 in HR) complete successfully.

Technical Reality

Devices within the same subnet communicate seamlessly via the switch's standard Layer 2 MAC address forwarding.
No SVI gateways are invoked for internal traffic.

The True Test: Validating Layer 3 Inter-VLAN Routing



Test Cases	Result
• Test 1: Finance (VLAN 40) successfully communicates with HR (VLAN 10). • | Test 2: IT (VLAN 20) successfully communicates with Sales (VLAN 30).	Cross-department communication confirmed. The switch successfully routes traffic between distinct subnets entirely in hardware.

The Anatomy of a Dropped Packet

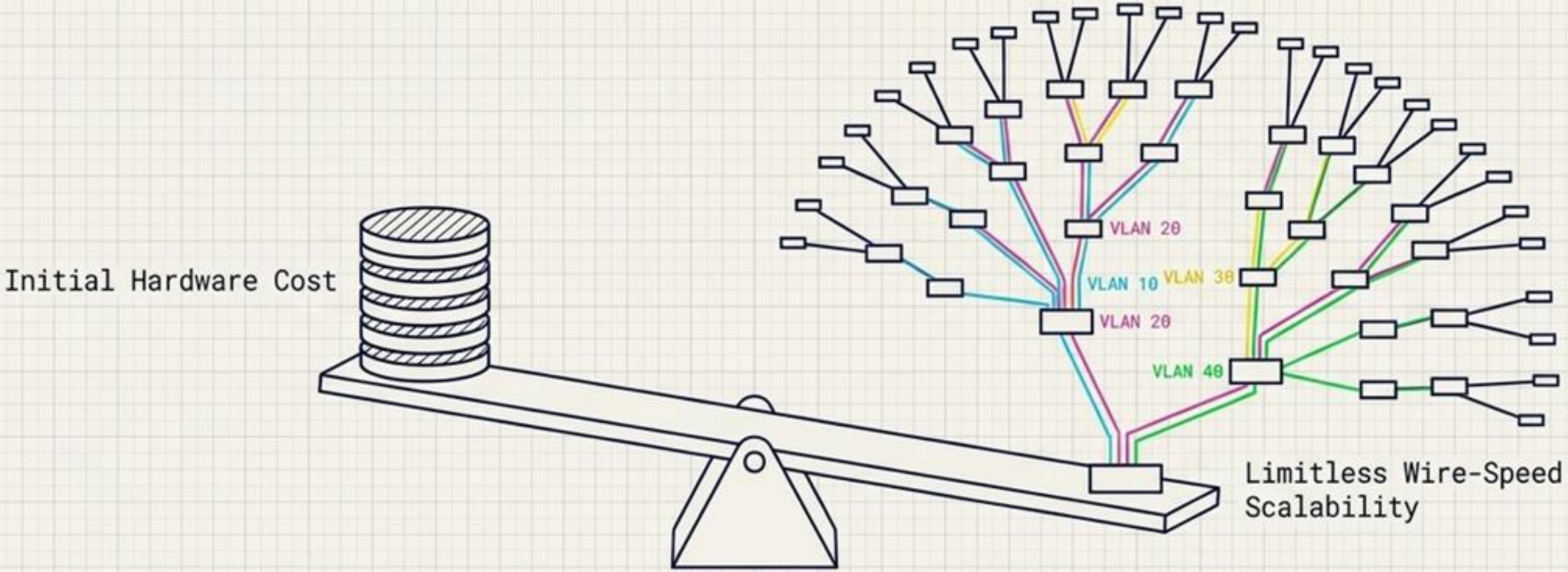


Context	Insight
During the initial cross-VLAN ping test, the first packet consistently drops.	This is a network feature, not a bug. The timeout reflects the initial Address Resolution Protocol (ARP) broadcast delay as the switch maps the new IP to a MAC address. Once learned, all subsequent wire-speed hardware routing succeeds instantly.

Architectural Synthesis & Comparison

	Traditional Routing	Router-on-a-Stick	Layer 3 Switch (SVI)
Hardware Required	2 Devices	2 Devices	1 Device
Scalability	 Poor	 Good	 Excellent
Throughput Speed	Standard	Bottlenecked	Wire-Speed Hardware
Cables per VLAN	1 Dedicated	Shared Trunk	Zero (Logical Only)

The Cost vs. Growth Trade-off



Analysis	The ROI
The primary trade-off of the SVI architecture is the higher initial price point of multilayer switches compared to standard Layer 2 switches.	Weighed against McNasir Systems' rapidly expanding budget and long-term departmental growth plans, the return on investment through simplified management, unbottlenecked performance, and eliminated router sprawl is definitive.

A Foundation for the Next Decade

Project Result

100% Inter-VLAN connectivity achieved across HR, IT, Sales, and Finance with zero external routing hardware.

Final Verdict

The migration from Traditional routing to Layer 3 Switching is complete. McNasir Systems is now equipped with a wire-speed, infinitely scalable network core ready to seamlessly absorb all future enterprise expansion.

